

6. Nitrate Vulnerable Zones (NVZ) are areas where nitrate concentrations are a risk to human health, or harmful to the aquatic environment.

Llangorse lake is the largest natural lake in south Wales. Approximately two-thirds of the surrounding land is used for intensive farming. Natural Resources Wales (NRW) has calculated that 99% of the nitrate found in Llangorse lake comes from intensive farming.

Table 6.1 is used by NRW to estimate the risk from nitrate levels in water.

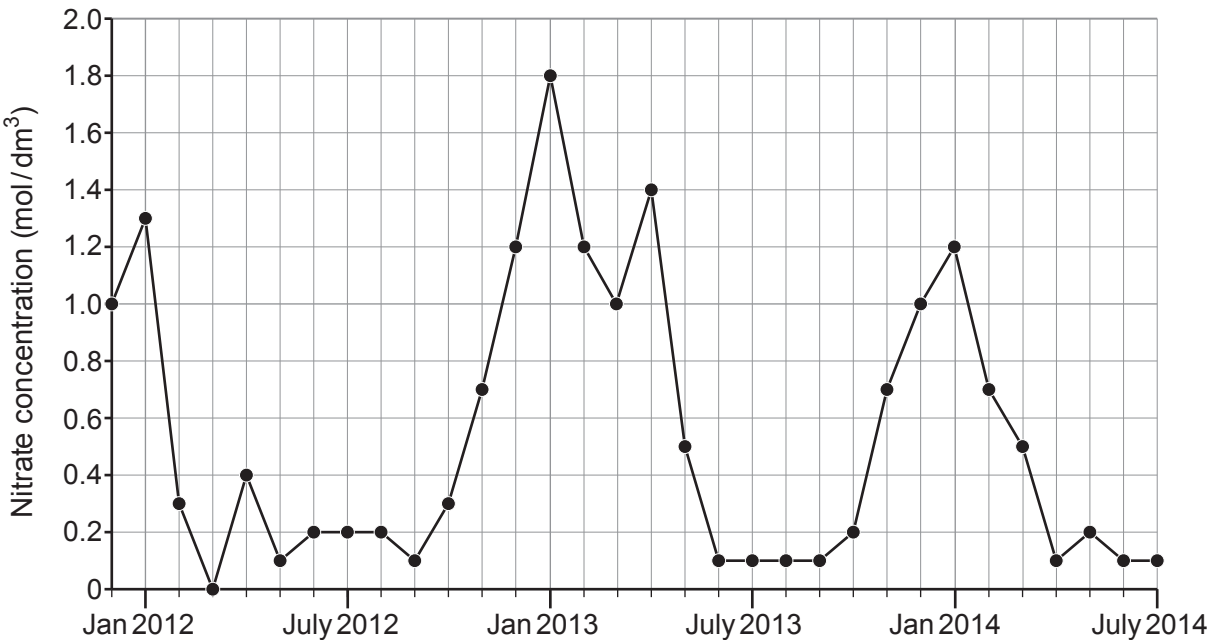
Table 6.1

Nitrate concentration (mol/dm ³)	Risk
< 0.5	very low
0.5 – 1.0	low
>1.0 – 1.7	low – medium
>1.7 – 3.5	medium
>3.5 – 6.0	medium – high
>6.0	high

Between December 2011 and July 2014, water samples were taken from Llangorse lake and the nitrate concentration analysed.

The results are shown in Graph 6.2.

Graph 6.2



Algal blooms were observed in Llangorse lake between April and October for each year shown in Graph 6.2.

- (a) (i) Using the information above, state the risk level for nitrates in Llangorse lake in January 2013.

[1]



Examiner
only

- (ii) Describe the annual trend common to all years shown in **Graph 6.2**. [2]

- (iii) Suggest a possible explanation for the common annual decrease in the nitrate levels in Llangorse lake. [2]

- (b) Explain how intensive farming leads to the formation of algal blooms and how these algal blooms can be harmful to aquatic animals. [6 QER]

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